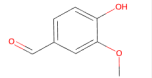


Graph Embedding Path

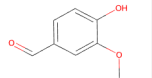


2D Molecular Graph

Graph Encoder
(GNN)

COc1cc(C=O)ccc1O
1D SMILES

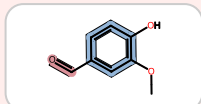
Is this molecule active in the SR-HSE assay?
Instruction



2D Molecular Graph

Source Predictor
(GNN)

Attribution



lowers

raises

Rationale Path

8 Learnable Queries

Projector (Q-Former)

Self-Attention

Cross-Attention

Self-Attention

x6

+ LoRA

Large Language
Model

Text
Output

Multi-granular Rationale

- ① Murcko sidechain C=O
- ② Murcko scaffold c1ccccc1
- ③ BRICS fragment O=Cc1ccc(O)cc1

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